



# East West University

## Department of CSE

### Mini Project Report

## “ Malware Family Classification Using a CNN with Attention on the MaleBin Dataset ”

**Course Code:** ICE 478/CSE475

**Section:** 2

**Submitted to:** Dr. Raihan Ul Islam, Associate Professor, Department of Computer Science and Engineering.

**Submitted by: Group-1**

| Student ID    | Student Name          |
|---------------|-----------------------|
| 2022-3-50-006 | Md. Mustafidur Rahman |
| 2022-3-50-008 | Amana Akter Arfa      |
| 2022-3-50-025 | Rakiba Alam           |

## Introduction

Malware detection has become an important research topic because cyber attacks are increasing every year. Traditional signature-based detection methods often fail to detect newly developed malware, which has encouraged researchers to use machine learning and deep learning techniques. In recent years, many researchers have converted malware binaries into grayscale images and applied image classification models for malware detection. CNN-based models, Vision Transformers, hybrid deep learning methods, and explainable AI have all shown promising performance. This section reviews five recent studies related to malware image classification and discusses their strengths and limitations.

## Related Work

| Ref | Paper (Year)                                                                                  | Dataset                                              | Method                                        | Attention / Special Module        | Best Result                               | Limitation                                                               |
|-----|-----------------------------------------------------------------------------------------------|------------------------------------------------------|-----------------------------------------------|-----------------------------------|-------------------------------------------|--------------------------------------------------------------------------|
| P1  | Self-Supervised Vision Transformers for Malware Detection (2022)                              | AndroZoo (1.2M apps, 47 malware types, 696 families) | Self-supervised Vision Transformer (SHERLOCK) | Vision Transformer Self-Attention | 97% binary accuracy                       | Requires very large unlabeled dataset and high computational resources.  |
| P2  | Grayscale Image-based Deep Learning Model for Efficient and Scalable Malware Detection (2025) | MalImg                                               | CNN on grayscale malware images               | No attention                      | High malware detection accuracy on MalImg | Only grayscale images; limited generalization to newer malware families. |

|    |                                                                                                |                                             |                                       |                                              |                        |                                                                           |
|----|------------------------------------------------------------------------------------------------|---------------------------------------------|---------------------------------------|----------------------------------------------|------------------------|---------------------------------------------------------------------------|
| P3 | Multimodal Malware Classification using Proposed Ensemble Deep Neural Network Framework (2025) | Android malware images + numerical features | Ensemble DNN + RUSBoost + Late Fusion | Feature Fusion (not attention)               | 95.36% accuracy        | Requires multiple data modalities and complex preprocessing.              |
| P4 | YoloMal-XAI: Interpretable Android Malware Classification Using RGB Images and YOLO11 (2025)   | CICMalDroid2020                             | YOLO11 + Explainable AI               | Grad-CAM, Grad-CAM ++, Eigen-CAM , HiRes-CAM | 99.87% binary accuracy | Uses RGB representation instead of grayscale; higher computational cost.  |
| P5 | Hybrid Noise-Resistant Technique for Malware Classification (2026)                             | Memory Forensics + BODMAS                   | CAE + LBP + XGBoost                   | Hybrid latent feature learning               | 99.90% accuracy        | Not an end-to-end CNN classifier; relies on handcrafted texture features. |

## **Discussion of Related Works**

### **Paper 1 (IEEE Access 2022)**

The authors proposed SHERLOCK, a self-supervised Vision Transformer for malware detection. Malware binaries were converted into grayscale images and processed using transformer self-attention. Unlike CNNs, Vision Transformers capture long-range dependencies between image patches. The model achieved 97% binary classification accuracy using over 1.2 million Android applications. However, the framework requires massive datasets and expensive training resources, making it difficult to deploy in ordinary academic environments.

### **Paper 2 (IEEE 2025)**

This study converts malware binaries into grayscale images and trains a CNN for malware classification using the MalImg dataset. The work demonstrates that image-based malware visualization enables automatic feature extraction without manual engineering. Although the model performs well on MalImg, it does not include attention mechanisms and its evaluation is limited to grayscale images, reducing its adaptability to more diverse malware datasets.

### **Paper 3 (Scientific Reports 2025)**

The authors proposed a multimodal malware classification framework combining malware images with handcrafted numerical features. A late-fusion architecture integrates an ensemble deep neural network with RUSBoost, achieving 95.36% accuracy. Although multimodal fusion improves robustness, the approach requires multiple feature sources and complicated preprocessing, increasing implementation complexity.

### **Paper 4 (JCP 2025)**

YoloMal-XAI transforms Android applications into RGB images and performs malware classification using YOLO11. The framework integrates explainable AI methods such as Grad-CAM and Eigen-CAM to visualize model decisions. It achieves 99.87% binary classification accuracy while providing high interpretability. Nevertheless, RGB conversion increases computational requirements and differs from grayscale malware visualization commonly used in malware image datasets such as MaleBin.

## Paper 5 (Scientific Reports 2026)

This paper introduces a hybrid framework combining Convolutional Autoencoders and Local Binary Pattern descriptors for malware visualization. Feature vectors are classified using XGBoost, producing 99.90% accuracy and demonstrating strong robustness under noisy conditions. Despite its excellent performance, the method depends on handcrafted texture descriptors rather than an end-to-end CNN architecture, making deployment more complicated.

## Comparative Analysis

| Feature            | Existing Works                            | Our Proposed Work        |
|--------------------|-------------------------------------------|--------------------------|
| Input              | Malware Images                            | MaleBin Grayscale Images |
| Feature Extraction | CNN / ViT / Autoencoder                   | CNN + Attention          |
| Attention          | Only some papers                          | Attention module         |
| Explainability     | Limited                                   | Grad-CAM (future work)   |
| End-to-End         | Some methods require handcrafted features | End-to-End CNN           |
| Dataset            | MalImg, CICMalDroid, AndroZoo, BODMAS     | MaleBin                  |
| Complexity         | Medium to Very High                       | Moderate                 |

|           |                        |                        |        |
|-----------|------------------------|------------------------|--------|
| Objective | Malware Classification | Malware Classification | Family |
|-----------|------------------------|------------------------|--------|

## Research Gap

Although recent studies have significantly improved malware image classification through CNNs, Vision Transformers, multimodal fusion, and hybrid feature extraction techniques, several important research gaps remain.

Most existing CNN-based approaches rely solely on convolutional feature extraction without incorporating attention mechanisms, which may limit the model's ability to focus on discriminative malware regions. Transformer-based models employ self-attention but generally require extremely large datasets and high computational resources, making them difficult to reproduce in academic environments. Multimodal frameworks improve classification accuracy by combining multiple data sources; however, they increase preprocessing complexity and require additional handcrafted features. Hybrid methods such as CAE-LBP combine learned and handcrafted features, sacrificing the simplicity of an end-to-end learning pipeline. Furthermore, many studies evaluate their models using datasets such as MalImg, CICMalDroid, AndroZoo, or BODMAS, while the MaleBin malware grayscale image dataset remains comparatively underexplored.

Therefore, this project proposes an end-to-end CNN with an Attention mechanism for malware image classification on the MaleBin dataset. The attention module is expected to emphasize informative regions within grayscale malware images while maintaining lower computational complexity than Vision Transformers and avoiding handcrafted feature engineering. This approach aims to provide a practical balance between classification performance, computational efficiency, and implementation simplicity for Track 3.

## References

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